

Association: Positive Control Analysis

2026-01-17

Methods

1. Mixscale (low MOI only)

Preprocessing: - Standard Seurat workflow: normalize data, find variable features, scale data, run PCA

Perturbation Signature Calculation (CalcPerturbSig):

- For each cell, find its 20 nearest non-targeting (NT) neighbors in PCA space with the same biological replicate
- Compute perturbation vector: $p_i = x_i - \text{Avg}(\text{neighbors})$
- This controls for biological replicate and systematic effects

Mixscale Scoring (RunMixscale):

- For each target, test all other genes (not the target itself) for differential expression
- If at least 5 genes are marginally DE, proceed; otherwise skip to simpler test
- Keep only those DE genes' measurements (at most 100 by default) in the perturbation vectors
- Compute perturbation scores by projecting onto the line from the perturbed cells' center to NT cells' center
- Compute standardized perturbation scores by z-transforming using the mean and std dev from NT cells
- NT cells centered around 0, perturbed cells "significantly" far away

Differential Expression Test (Run_wmvRegDE):

- For each target-gene pair that got Mixscale scores, run negative binomial regression:
- Response: gene expression values
- Predictors: perturbation scores (0 for NT cells, standardized score for perturbed cells) + log10 library size
- The p-value comes from testing whether the coefficient on the perturbation score is significant
- This is a weighted DE test where the weight reflects how perturbed each cell is

Output:

- Returns p-values for each target-gene pair (currently all pairs, though on-target only would be more efficient)
2. FR-Perturb
 - We compare two versions: using their skew-t approximation, or resampling with 2000 samples.
 3. SCEPTRE

TODO

Datasets:

Dataset name: `replogle-rd7_large`

Number of cells: 210110 ; number of genes: 1619 ; number of unique guides: 1813

Dataset name: `gasperini_pos_control`

Number of cells: 185171 ; number of genes: 377 ; number of unique guides: 855

1. Subsampled Replogle

- run `sceptre-pipeline` on the full Replogle-rd7. `method = mixture`, with the default formula used, so we regressed on `response_n_nonzero`, `response_n_umis`, `grna_n_nonzero`, and `grna_n_umis` (all log or log1p as needed). `batch` was not used because it has 56 levels.

Then, to get the desired subset of the response matrix:

- Filter to low MOI cells: Keep only cells expressing exactly one gRNA
 - Select positive control cells:
 - * Iterate through response genes in dataset order
 - * For each gene, collect all cells expressing gRNAs targeting that gene
 - * Skip genes with fewer than `min_num_cells_per_target` cells (default: 20, used for “large”)
 - * Continue until reaching `num_pc_cells` positive control cells (e.g., 200,000 for “large”)
 - Add non-targeting cells: Include cells expressing non-targeting gRNAs up to `num_nt_cells` (e.g., 10,000 for “large”), up to all of them
 - Extract expression data: Subset the response matrix to selected genes and cells
 - Quality filtering: Remove cells with zero expression across all genes, then remove any gRNAs left with zero cells
 - Export in multiple formats: Write data for `sceptre` (binary indicator matrix), `mixscale` (named vectors), and `FR-Perturb` (h5ad)

The resulting `response_matrix` resembles a block diagonal one, with ambient noise off the diagonal, as the first gene’s targeting cells are the first cells in, and so on, until the final columns of NT cells.

2. Subsampled Gasperini

- Run `sceptre-pipeline` on the full Gasperini data, `method = mixture`, with the default formula used, so we regressed on `response_n_nonzero`, `response_n_umis`, `grna_n_nonzero`, `grna_n_umis` (all log or log1p as needed), and `batch` (2 levels)
- Identify on-target genes: Find genes that are both targeted by gRNAs and present in the response matrix
 - Collect cells with on-target gRNAs:
 - * For each on-target gene, get all gRNAs targeting that gene
 - * Collect all cells expressing any of these gRNAs (no filtering to single gRNA per cell - this is high MOI)
 - Add non-targeting cells: Include all cells expressing non-targeting gRNAs
 - Create cell-level assignments: Build long-form dataframe mapping each cell to its expressed gRNAs (cells can have multiple gRNAs)
 - Extract expression data: Subset response matrix to on-target genes and selected cells
 - Compute gRNA metrics: Calculate `grna_n_nonzero_subset` and `grna_n_umis_subset` from the gRNA UMI counts for the cell subset
 - Export in multiple formats:

- * sceptre: Binary indicator matrix, cell covariates including prep_batch
- * FR-Perturb: h5ad file with colon-delimited multi-perturbation column (e.g., “GENE1:GENE2” for cells with multiple gRNAs), includes prep_batch covariate

Beginning with some tables:

Table 1: Summary statistics by dataset and method. The column ‘Corr. with n’ gives the Spearman correlation of the negative p-value with the number of cells expressing each target. ‘Pcnt. ≤ 0.05 ’ gives the percentage of p-values at or below 0.05. Both datasets are downsampled to only have on-targets.

Dataset	Method	Num. Targets	Mean p-value	Corr. with n	Pcnt. ≤ 0.05
Gasperi	FR-Perturb (resamp)	377	0.2510	0.031	43%
Gasperi	FR-Perturb (skew)	377	0.2530	0.022	43.5%
Gasperi	SCEPTRE	377	0.0113	0.177	98.1%
Replogle	FR-Perturb (resamp)	1619	0.2850	0.068	33.5%
Replogle	FR-Perturb (skew)	1619	0.2840	0.067	34.1%
Replogle	Mixscale	1619	0.1650	0.313	78.1%
Replogle	SCEPTRE	1619	0.0456	0.318	87.7%

Table 2: Pairwise method comparisons within each dataset. ‘Pcnt. $1 \leq 2$ ’ gives the percentage of times the p-value for Method 1 is at or below the p-value for Method 2. ‘Corr.’ gives the Spearman correlation between the p-values for Methods 1 and 2. Both datasets are downsampled to only have on-targets.

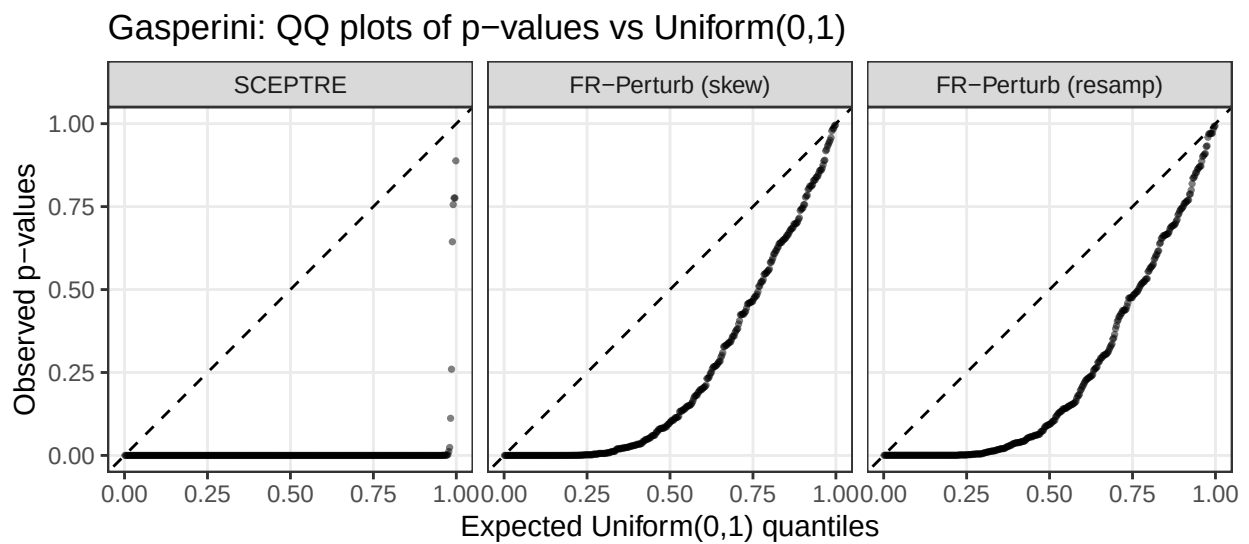
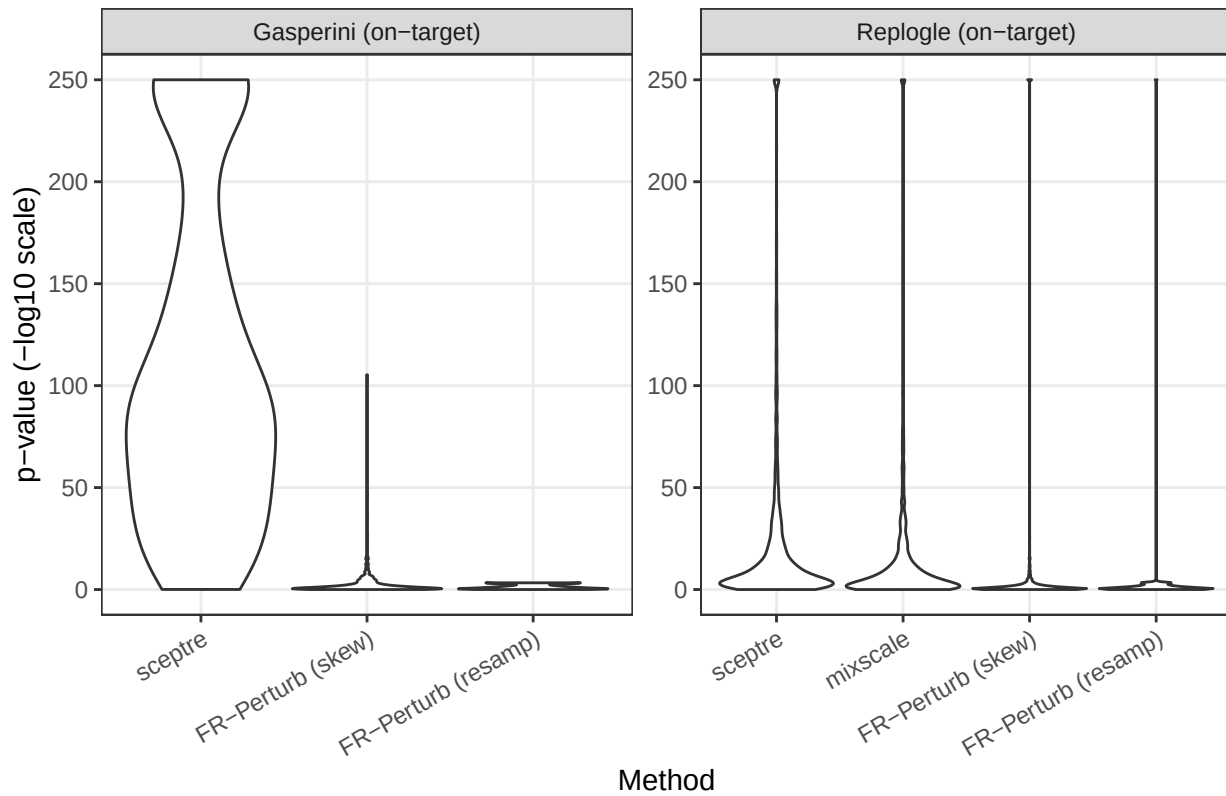
Dataset	Method 1	Method 2	Pcnt. $1 \leq 2$	Corr.
Gasperi	FR-Perturb (skew)	FR-Perturb (resamp)	59.4%	0.990
Gasperi	SCEPTRE	FR-Perturb (resamp)	98.4%	0.233
Gasperi	SCEPTRE	FR-Perturb (skew)	98.1%	0.238
Replogle	FR-Perturb (skew)	FR-Perturb (resamp)	55.8%	0.996
Replogle	Mixscale	FR-Perturb (resamp)	77.9%	0.007
Replogle	Mixscale	FR-Perturb (skew)	76%	0.005
Replogle	SCEPTRE	FR-Perturb (resamp)	89%	0.014
Replogle	SCEPTRE	FR-Perturb (skew)	86.8%	0.013
Replogle	SCEPTRE	Mixscale	81.1%	0.926

Visualization of on-target p-values

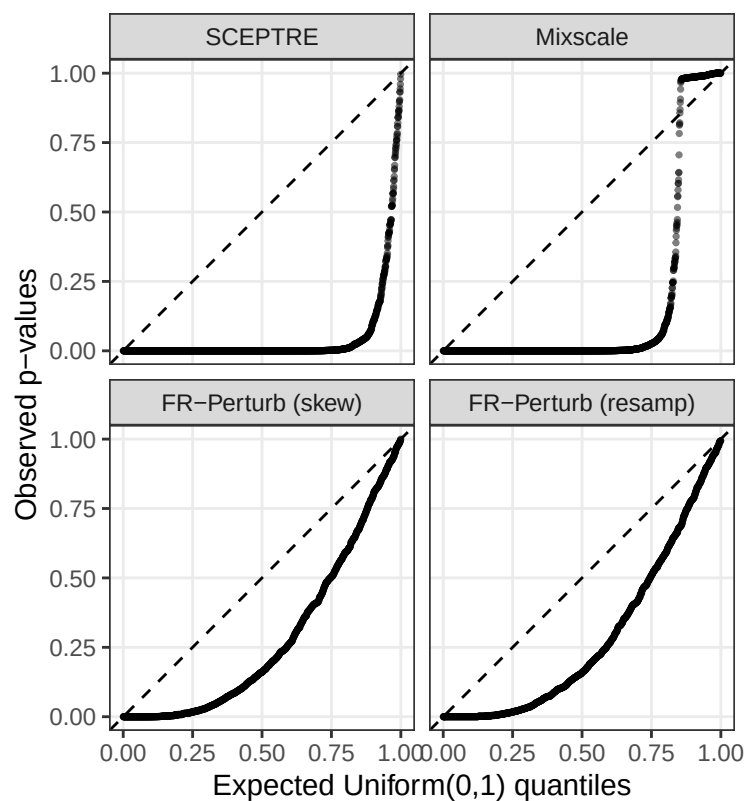
We begin with violin plots to show the marginal distribution on the $-\log_{10}$ scale. Then QQ plots give more insight.

To begin with, we simply show the marginal distributions of the p-values side by side.

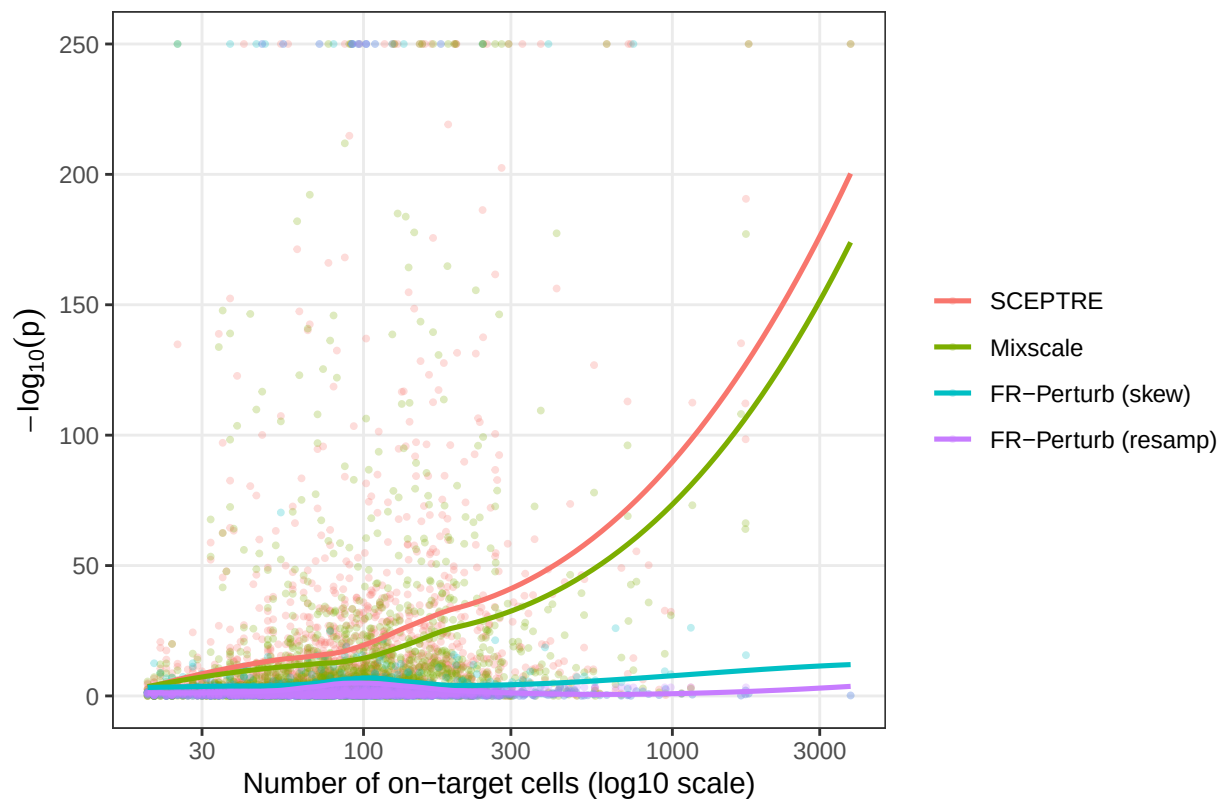
Comparison of on-target p-values



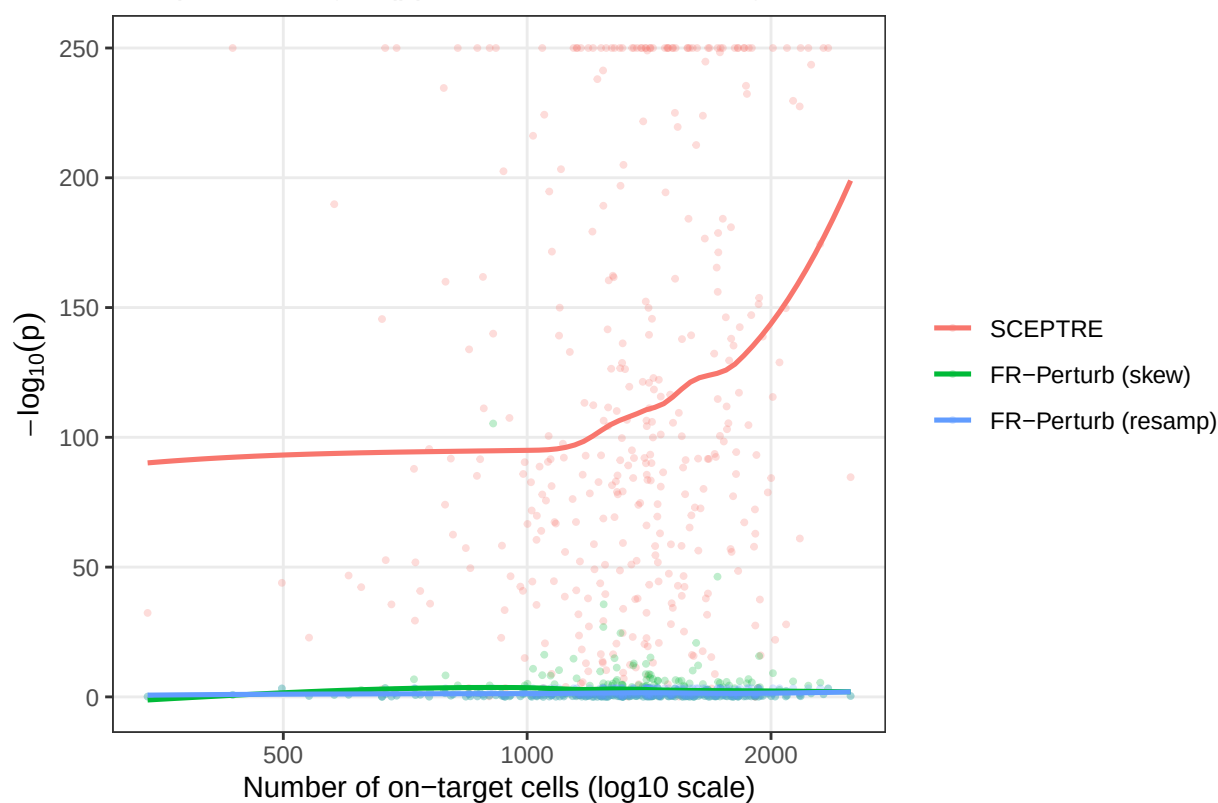
Replogle: QQ plots of p-values vs Uniform(0,1)



Replogle: $-\log_{10}(p)$ vs number of on-target cells

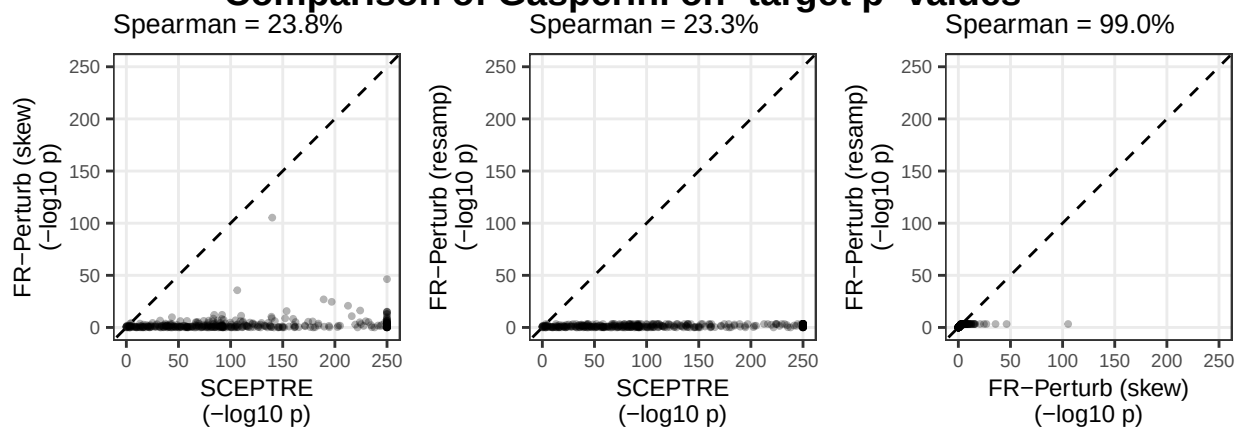


Gasperini: $-\log_{10}(p)$ vs number of on-target cells

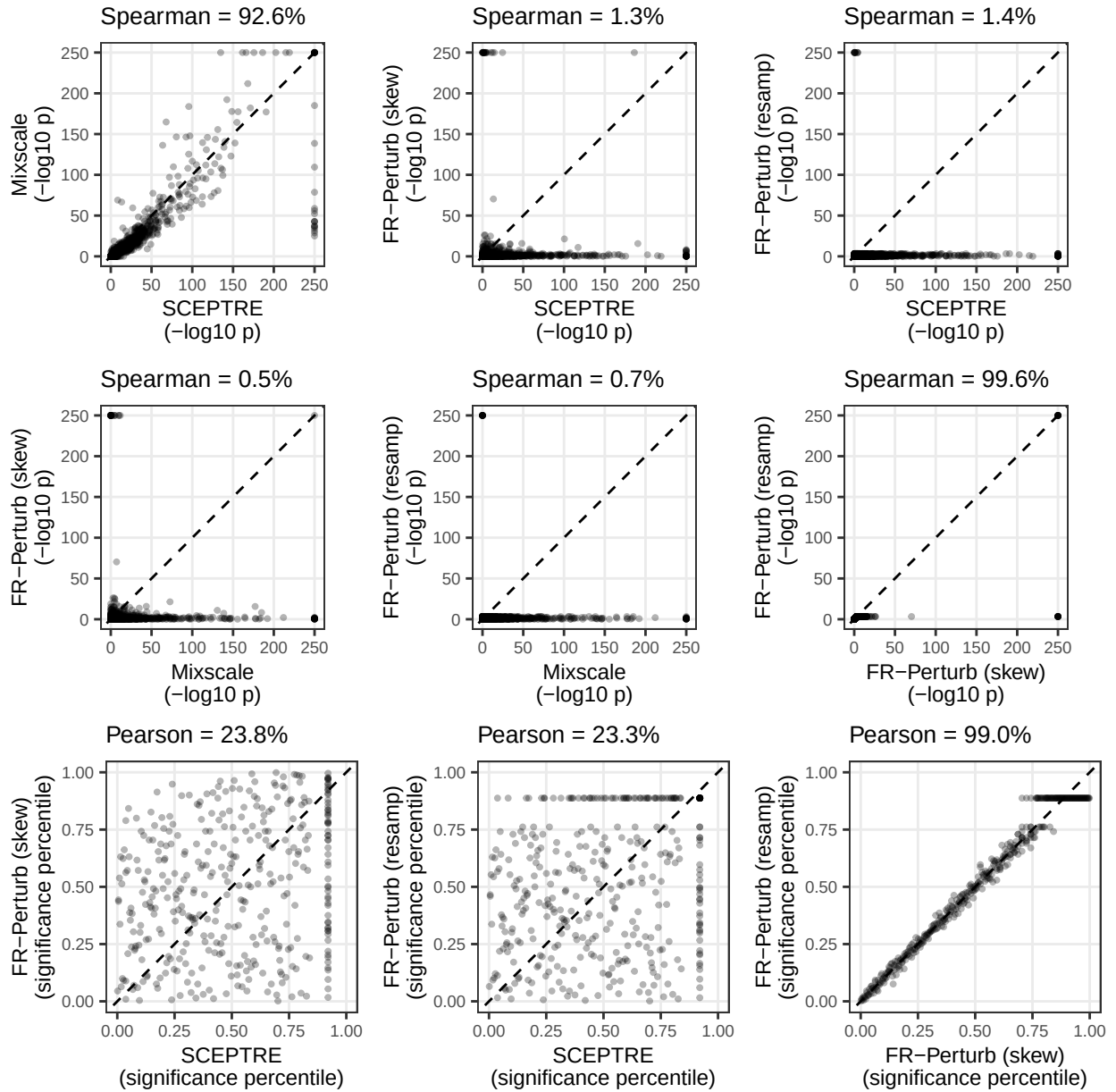


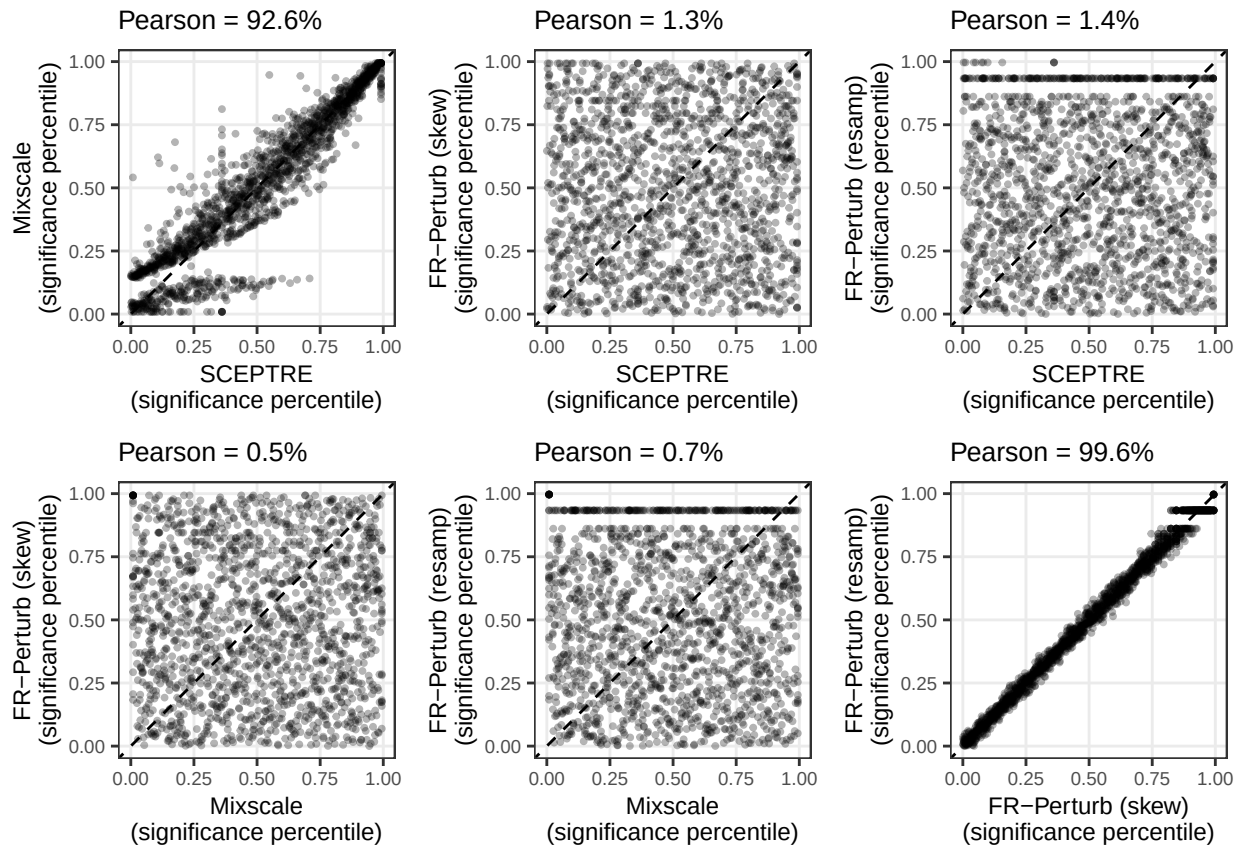
Next, we want to explore the extent to which these p-values agree.

Comparison of Gasperini on-target p-values



Comparison of Replogle on-target p-values





plot: i want to indicate the sample size somehow

TODO

- More methods?
- More datasets?
- Connect to guide assignment? Try with different assignments? Is it circular to use sceptre's for sceptre?
- Use covariates like total sequencing depth for guides or genes in the p-value assessment